Subh Aush Singh

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Experience

Qbayes **Feb. 2026 - Mar. 2026**

Full Stack Developer Intern *Remote*

- Architected a **production-grade multi-tenant ERP** backend for a bakery chain with strict **data isolation** across companies, locations, and shops using **Node.js, Express, PostgreSQL, and Prisma ORM**.
- Implemented **hierarchical RBAC** across 5 roles with role-scoped Prisma queries, enforced at the service layer (not just routes), alongside **JWT auth** with **HttpOnly cookies**, refresh token rotation, and **multi-device session management**.
- Built stock lifecycle management (inbound → distribution → sales → expiry returns), **atomic Prisma transactions** for every multi-step operation, and **node-cron** jobs for automated expiry alerts at 7, 3, and 0 days.

Freelance **2026**

Frontend Developer *Remote*

- Designed and developed a **responsive, production-ready landing page** for Aston Glass to establish their digital presence and closely align with client branding guidelines.
- Engineered the frontend for rapid load times and performance, achieving an optimized **Largest Contentful Paint (LCP) of 0.9s**.
- Attained **Lighthouse scores of 96 Accessibility, 96 Best Practices, 92 SEO, and 84 Performance** through **semantic HTML** and efficient asset delivery.

Projects

TCP HTTP Server | *Node.js, Raw TCP Sockets, Playwright* | GitHub **2026**

- Engineered a **zero-dependency HTTP/1.1 web server** from scratch using Node's **raw `net` module**, implementing a **dynamic buffer** for connection handling, manual request/header parsing, and **chunked transfer-encoding**.
- Implemented the complete **RFC 6455 WebSocket protocol** directly over TCP, successfully handling the **Upgrade handshake**, frame parsing, payload **masking**, ping/pong control frames, and **message fragmentation**.
- Developed static file serving with **206 Partial Content** (Range) and **304 Not Modified** caching support, alongside on-the-fly **gzip compression** and a comprehensive **Playwright test suite**.

Credex Audit | **AI Spend Diagnostic Tool** | *Next.js, Express, Supabase, Groq, Zod* | Live **2026**

- Built a diagnostic tool that flags redundant AI subscriptions and recommends tier downgrades; shipped a **decoupled Express API** (custom **rate-limiting**, **HPP protection**) instead of Next.js API routes for clean process control.
- Achieved **Lighthouse scores of 96+**; implemented **fire-and-forget lead capture** (Supabase + Resend) so DB/email latency never blocks the user-facing response.

Open Source Contributions

strapi/strapi | PR #26421 (Merged)

- **Guarded a repeatable-field `.map()` crash** on relations in the content-manager package.

emscripten-core/emscripten | PR #26639

- Allowed **overriding `-jsD` directives** from the command line.

emscripten-core/emscripten | PR #26318 (Merged)

- **Fixed a missing `$webSockets` dependency** in `emscripten_websocket_new`; added a **regression test** reviewed and approved by core maintainers kripken and sbc100.

CollaboraOnline/online | PR #14444 (Merged)

- **Rewrote `create-l10n-all-js.pl`** in **Python** for the localization build pipeline.

marketcalls/openalgo | PR #898 (Merged)

- **Added precise type hints** to trading API helper functions in `restx_api/ticker.py`, improving **type safety** and IDE tooling.

nextcloud/server | PR #58370 (Merged)

- **Removed a redundant `ext-json` Composer requirement**, since JSON is bundled with PHP 8+.

Technical Skills

Languages: C/C++, Python, JavaScript, TypeScript
Frameworks & Libraries: React.js, Next.js, Node.js, Express, TailwindCSS, LangChain
Databases: MongoDB, PostgreSQL
Tools & Platforms: Git & GitHub, Prisma, Inngest, Zod, Supabase
Problem Solving: 160+ problems solved on LeetCode

Education

Techno Main Salt Lake **Sep. 2023 - May 2027**

B.Tech in Computer Science & Engineering | 79% *Kolkata, West Bengal*

Achievements

Xplore | Career Counselling Platform | SIH 2024 (College-Level), 2nd Place **2024**

- Built for the SIH problem statement on career counselling in schools, developing a **full-stack platform** (web, mobile, ML) that recommends careers via a **test-driven ML model** and surfaces free learning resources.